

EXPLORATORY DATA ANALYSIS DASHBOARD USING EXCEL & POWER BI

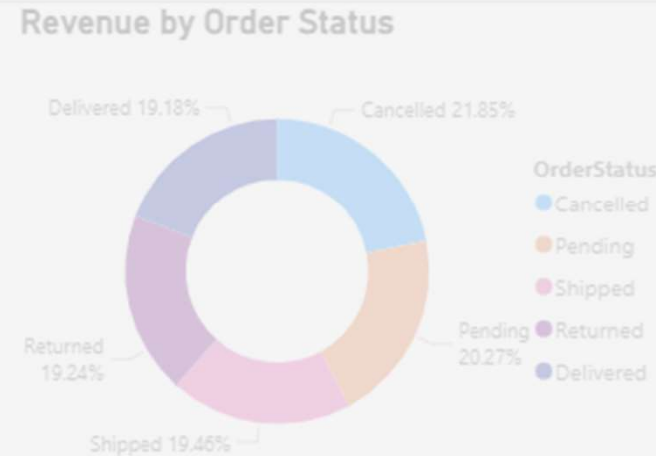
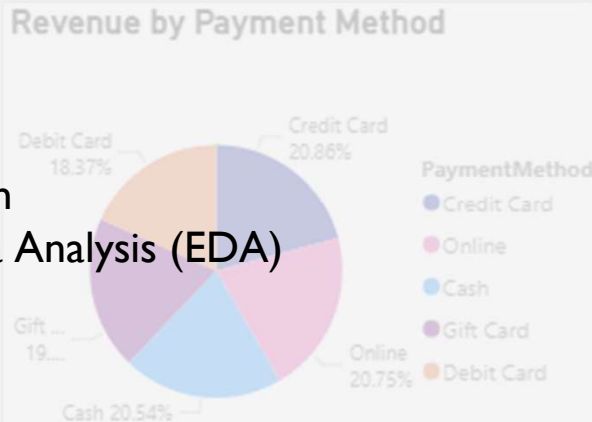
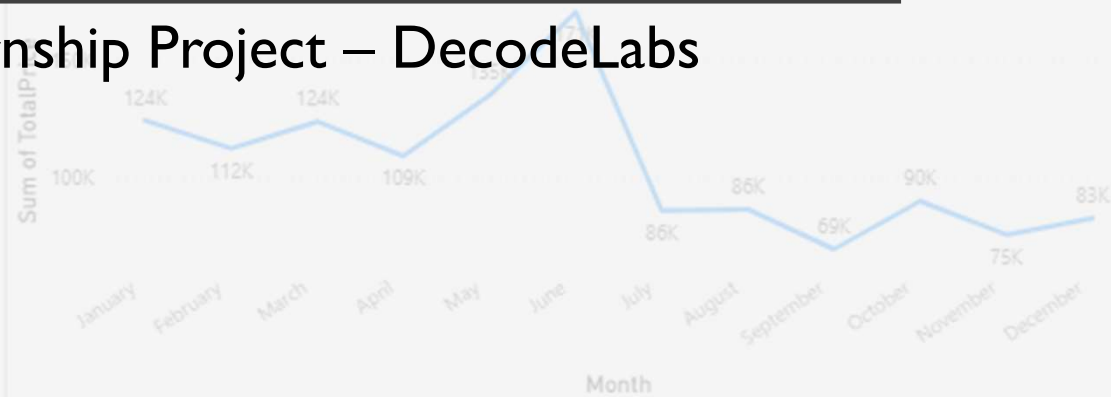
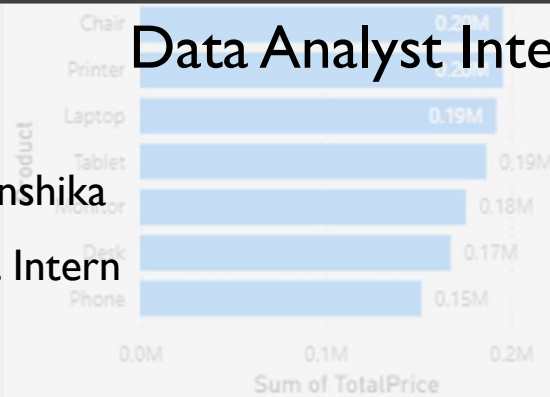
Data Analyst Internship Project – DecodeLabs

Prepared By: Vanshika

Role: Data Analyst Intern

Tools Used:

- Microsoft Excel
- Power BI
- Data Cleaning
- Data Visualization
- Exploratory Data Analysis (EDA)



KEY INSIGHTS

- ✓ Total Revenue shows an increasing trend with 18.6% growth compared to the previous period.
- ✓ Laptop and Printer contribute the highest revenue among all products.
- ✓ Online and Credit Card are the most preferred payment methods.
- ✓ Cancelled orders have the highest share among order statuses.

Project Overview

As part of my Data Analyst Internship at DecodeLabs, I worked on an end-to-end data analytics project involving data cleaning, preprocessing, statistical analysis, exploratory data analysis, and dashboard visualization using Excel and Power BI.

The project was divided into two major tasks:

Project 1 – Data Cleaning & Preprocessing

The dataset was cleaned and organized using Microsoft Excel to improve data quality and prepare it for analysis.

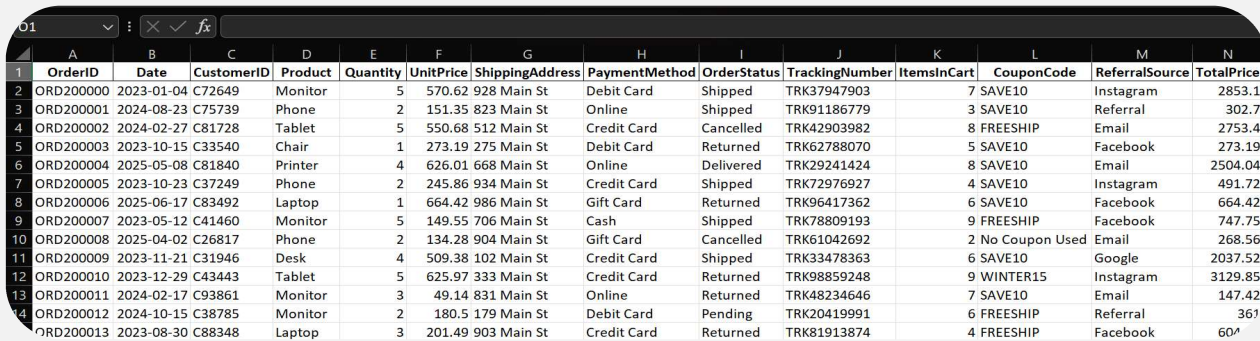
Project 2 – Exploratory Data Analysis & Dashboard Visualization

Exploratory Data Analysis (EDA) was performed to identify trends, patterns, and business insights from the dataset. An interactive Power BI dashboard was then created to visualize important business metrics such as revenue trends, product performance, payment methods, and order status distribution.

This project helped in understanding how raw business data can be transformed into actionable insights using analytical and visualization techniques.

Data Cleaning & Preprocessing

As part of Project I, the dataset provided by DecodeLabs was cleaned and prepared using Microsoft Excel before visualization and analysis.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
2	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
3	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
4	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
5	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
6	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
7	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
8	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
9	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
10	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon Used	Email	268.56
11	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
12	ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
13	ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
14	ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
15	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604

The following preprocessing steps were performed:

- ✓ Removed duplicate records
- ✓ Checked for missing values
- ✓ Corrected inconsistent formatting
- ✓ Organized column structure
- ✓ Verified data types for analysis

Note:

Only a small portion of the dataset is displayed for demonstration purposes to maintain data confidentiality and professional standards.

Exploratory Data Analysis (EDA)

As part of Project 2, Exploratory Data Analysis (EDA) was performed to analyze the dataset and identify meaningful business insights before dashboard creation.

Basic statistical measures were calculated to better understand the numerical behavior of the dataset and support the exploratory analysis process.

The statistical analysis included:

- Mean (Average)
- Median
- Count of Records

The analysis was performed on important numerical columns such as:

- TotalPrice
- Quantity
- ItemsInCart

These statistical measures helped in understanding customer purchasing behavior and overall business trends.

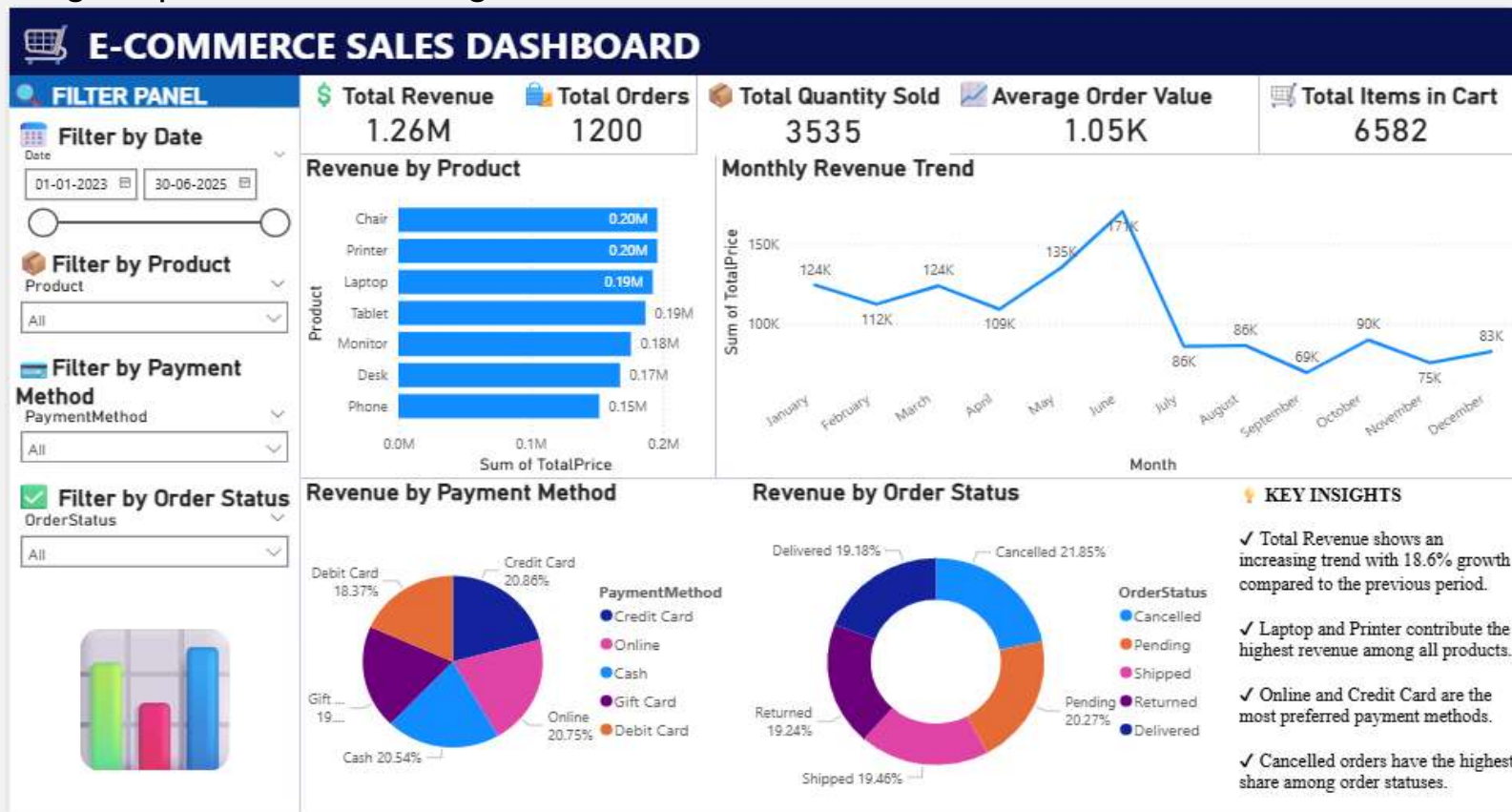
	A	B	C	D
1	Metric ▾	Mean ▾	Media ▾	Coun ▾
2	TotalPrice	1053.9683	823.615	1200
3	Quantity	2.94583333	3	1200
4	ItemsInCart	5.485	5	1200



Interactive Power BI Dashboard

After completing the Exploratory Data Analysis process, an interactive Power BI dashboard was developed to visualize and analyze the cleaned dataset effectively.

The dashboard was designed to convert raw business data into meaningful visual insights for better decision-making and performance tracking.



Dashboard Features:

- KPI Cards
- Revenue Analysis
- Product Performance Tracking
- Monthly Revenue Trends
- Payment Method Analysis
- Order Status Analysis
- Interactive Slicers & Filters

The dashboard provides a clear and interactive representation of business performance, customer purchasing patterns, and sales trends.

Key Insights from Project-2

- ✓ Top-performing products generated the highest share of revenue.
- ✓ Revenue trends varied across different months, helping identify peak business periods.
- ✓ Digital payment methods were more frequently preferred by customers.
- ✓ Delivered orders contributed the highest revenue compared to other order statuses.
- ✓ Interactive slicers improved dashboard usability and data exploration.
- ✓ KPI cards provided quick visibility into overall business performance.
- ✓ Exploratory Data Analysis helped identify important sales and customer behavior patterns.

Conclusion

This internship project at **DecodeLabs** helped strengthen my understanding of both **Project 1 (Data Cleaning & Preprocessing)** and **Project 2 (Exploratory Data Analysis & Dashboard Visualization)** using Excel and Power BI.

Through this project, I gained practical experience in:

- Cleaning and preprocessing raw datasets
- Performing exploratory data analysis
- Calculating statistical measures
- Creating interactive dashboards
- Generating business insights through visualization

Overall, this project improved my analytical thinking, dashboard design skills, and understanding of business intelligence reporting.